

DATA VORTEX A'26 — ROUND 3

Real-Time Social Reaction Analysis Notebook

Assigned Topic: Public Reaction to a Platform Monetisation Change

Platform/Event Focus: X (Twitter) monetisation and the transition toward Original Content Rewards

Analysis Dataset: 40 manually reviewed public comments collected from YouTube videos discussing X monetisation changes.

1. Dataset Overview

Metric	Value
Final comments	40
Unique videos	12
Duplicate comments	0
Earliest comment	2026-08-05 09:02:24+00:00
Latest comment	2026-09-10 22:08:42+00:00
Total comment engagement	94
Maximum comment engagement	15

The raw collection contained 326 comments. After platform/topic filtering, comment-quality review, and manual relevance review, 40 comments were retained for the final analytical dataset. The original raw dataset was preserved separately.

2. Data Collection Method

Public YouTube comments were collected using the YouTube Data API v3. Search queries focused on X monetisation, X creator monetisation, X creator revenue sharing, X monetisation in 2026, and Original Content Rewards. Video statistics and public comment metadata were collected together with timestamps and engagement counts.

Raw collection: 326 comments from 39 videos.

Final analytical dataset: 40 comments from 12 videos.

3. Preprocessing and Quality Control

- Missing-value audit was performed.
- Duplicate comment IDs were checked.
- Clearly unrelated YouTube/Facebook and other unrelated videos were removed.
- Comments unrelated to the monetisation discussion were excluded during manual review.
- Extremely low-information and purely generic comments were reviewed.
- The raw collection was preserved and filtering was performed on derived datasets.

- The final dataset was sorted chronologically for time-based analysis.

4. NLP Model Application

The sentiment predictions were generated using the DistilBERT model developed and evaluated during Round 2. The locally saved Round 2 model was loaded together with its tokenizer and applied to each final Round 3 comment.

Model	DistilBERT
Task	3-class sentiment classification
Labels	Negative / Neutral / Positive
Maximum sequence length	128 tokens
Round 3 comments analyzed	40
Average prediction confidence	0.6760

Important limitation: Model predictions are treated as model outputs rather than manually verified ground truth. This is especially important for short, multilingual, slang-heavy, or context-dependent comments.

5. Sentiment Distribution

Sentiment	Comments	Percentage
Negative	7	17.5%
Neutral	26	65.0%
Positive	7	17.5%

Neutral predictions form the largest category in the collected sample. Positive and negative predictions are present in smaller numbers. This indicates that the observed discussion contains substantial informational and question-oriented content in addition to explicit positive or negative reactions.

6. Time-Based Sentiment Analysis

Daily sentiment proportions were calculated from the timestamped comments. A sentiment balance was defined as Positive percentage minus Negative percentage.

Date	Negative	Neutral	Positive	Total	Balance
2026-08-05	0	1	0	1	0.00
2026-08-08	0	5	4	9	44.44
2026-08-09	0	3	1	4	25.00
2026-08-10	1	0	1	2	0.00
2026-08-11	1	1	0	2	-50.00
2026-08-12	1	0	1	2	0.00
2026-08-13	1	0	0	1	-100.00
2026-08-14	0	3	0	3	0.00
2026-08-19	0	2	0	2	0.00
2026-08-25	0	1	0	1	0.00
2026-08-31	0	2	0	2	0.00
2026-09-02	1	0	0	1	-100.00
2026-09-04	0	1	0	1	0.00
2026-09-05	1	0	0	1	-100.00

Date	Negative	Neutral	Positive	Total	Balance
2026-09-06	0	1	0	1	0.00
2026-09-10	1	6	0	7	-14.29

Observed Shift 1: The Aug 8–Aug 13 period moves from a positive/mixed balance toward increasingly negative or concern-oriented observations. The later dates in this sequence have small sample sizes, so this should be interpreted as an observed shift in the collected comments rather than a population-level statistical conclusion.

Observed Shift 2: Between Sep 6 and Sep 10, the sample moves from one neutral comment to a larger Sep 10 sample dominated by neutral/question-oriented comments with one negative prediction. This represents a weaker shift toward concern/question-oriented discussion.

7. Engagement Spike Analysis

Mean comment engagement was 2.35, median engagement was 1.50, and the standard deviation was 2.98. An engagement spike was defined as engagement greater than or equal to mean + 2 standard deviations.

Date	Engagement	Video	Comment
2026-08-08 15:17	15	X Monetization Just Changed... Again!	I've been waiting for your video on this Sir
2026-08-08 18:33	10	X Monetization Just Changed... Again!	monetized creators now will get 4-5x or even 6x the payout amount

Both detected spikes occurred on Aug 8 under the video "X Monetization Just Changed... Again!". The reactions focused on the newly discussed monetisation change, creator expectations, and payout implications. The payout multiple mentioned in one comment is treated as a commenter claim rather than independently verified fact.

8. Topic and Entity Analysis

Category	Topic / Entity	Comments	Share
Topic	Eligibility & Access	11	27.5%
Topic	Original Content & Reposts	10	25.0%
Topic	Monetisation Rules	6	15.0%
Topic	Payouts & Earnings	5	12.5%
Topic	Impressions & Reach	5	12.5%
Topic	Small Creators	4	10.0%
Entity	X / Twitter	10	25.0%
Entity	Pakistan	6	15.0%
Entity	500K Impressions	3	7.5%
Entity	Original Content Rewards	2	5.0%
Entity	Elon Musk	1	2.5%

The most frequent detected topics were Eligibility & Access and Original Content & Reposts. Other recurring areas included monetisation rules, payouts/earnings, impressions/reach, and small creators.

Topic percentages are keyword-based and are not mutually exclusive; one comment can contribute to more than one topic.

9. Trigger Explanations

Monetisation-rule change: The strongest concentration of engagement occurred around videos discussing the change to X monetisation.

Eligibility and access: Several comments focused on country eligibility, verification, account access, and whether particular creator/account types could participate.

Original content and reposts: Comments discussed original content, repost-related eligibility, and how creators should adapt their content.

Impressions and creator thresholds: Multiple comments asked how to reach or satisfy impression-related requirements.

10. Limitations

- The dataset consists of public YouTube comments, not a complete sample of all X users.
- YouTube search results can contain videos with mixed or indirect relevance, requiring filtering.
- The final analytical sample contains only 40 comments.
- Several dates have very small numbers of comments, limiting the strength of daily sentiment comparisons.
- The Round 2 DistilBERT model was trained on a different labelled dataset and may be less reliable for multilingual slang-heavy comments.
- Keyword-based topic/entity analysis can miss implicit references and may assign multiple categories to one comment.
- Commenter statements about payouts, eligibility, or platform rules are observations or claims and were not independently treated as verified facts.

11. Analytical Conclusion

Within this collected sample, discussion around the X monetisation change was concentrated on eligibility and access, original-content/repost rules, monetisation requirements, payouts, and impressions. The largest engagement spikes occurred on Aug 8 around a video specifically discussing the monetisation change. Sentiment predictions were predominantly neutral, with smaller positive and negative groups. The observed temporal movements provide evidence of changing discussion patterns within the collected sample, while the small and platform-specific sample limits generalisation to the wider public.